

Assignment: 03

Working with LLM APIs and Parameters

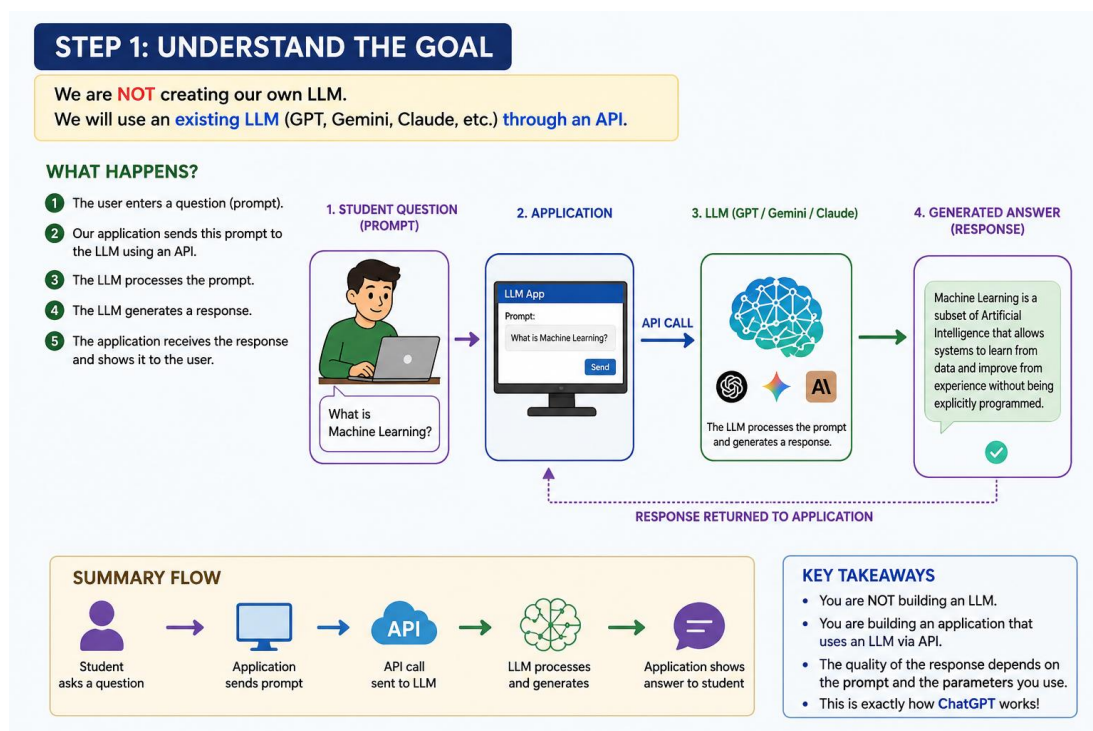
Problem Statement

Develop an application that integrates an LLM API to generate responses.

The system should allow modification of parameters such as **Temperature** and **Maximum Tokens**.

Analyze the impact of these parameters on response quality, creativity, and length, and report the findings.

Step 1: Understand the Goal



Step 2: What is an API?

API stands for

Application Programming Interface

STEP 2: WHAT IS AN API?

API stands for **Application Programming Interface**.
It is a messenger that allows your application to talk to an LLM.
Your application does not talk to the model directly.
It communicates **through the API**.

THINK OF IT LIKE ORDERING FOOD



You interact with the waiter, not the kitchen. Similarly, you interact with the API, not the LLM directly.

EXAMPLE



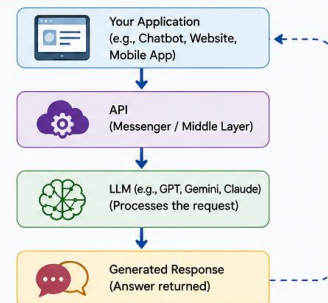
KEY TAKEAWAY

- Your application never talks directly to the LLM.
- It talks to the API.
- The API understands your request, sends it to the LLM, gets the response, and returns it to your application.
- This makes integration **simple, secure and scalable**.

Definition

An API (Application Programming Interface) is a set of rules and protocols that allows different software systems to communicate with each other.
In simple words, API is a **bridge** between your application and the LLM.

HOW IT WORKS WITH LLM



Response travels back the same way through the API to your application.

REAL-WORLD EXAMPLES



WHY API IS IMPORTANT?

- ✓ Keeps the system secure.
- ✓ Handles complex communication.
- ✓ Makes it easy to switch models.
- ✓ Supports scale and reliability.

Step 3: What students will build?

A simple application.

User enters

Explain Artificial Intelligence

Application sends it to GPT/Gemini.

Model returns

Artificial Intelligence is the simulation of human intelligence...

Application displays it.

Very simple.

Step 4: Understand API Parameters

STEP 4: UNDERSTAND API PARAMETERS

- When we call an API, we don't just send the prompt. We also send **parameters**.
- These parameters control how the model generates the response.

EXAMPLE (Python)

```
response = model.generate_content(  
    prompt="Explain AI",  
    temperature=0.8,  
    max_tokens=300  
)
```

- The question or instruction we send to the model
- Controls creativity / randomness of the response
- Controls the maximum length of the response

HOW THE API CALL WORKS



1. Your Application
You enter the prompt and set parameters.



2. API
Your application sends the prompt + parameters to the API.



3. LLM (Model)
The model processes the prompt using the given parameters.



4. Response
The API returns the generated response to your application.

COMMON PARAMETERS (You will learn more in next steps)

Parameter	What it does	Example Values
temperature	Controls creativity. Higher = more creative, Lower = more predictable.	0 (very factual) 0.5 (balanced) 1.0 (very creative)
max_tokens	Controls the maximum length of the response.	50, 100, 300, 500, 1000
(others)	Top_p, Frequency_penalty, Presence_penalty, etc.	Advanced control over the output

EXAMPLE API REQUEST

You send:

```
{  
  "prompt": "Explain AI",  
  "temperature": 0.8,  
  "max_tokens": 300  
}
```

- The question we ask
- Temperature = 0.8 → balanced creativity and accuracy
- Maximum length of response (up to ~300 tokens)



Model returns:

"Artificial Intelligence (AI) is the simulation of human intelligence in machines that are programmed to think and learn. It includes technologies like machine learning, natural language processing, computer vision, and robotics ..."
(response up to ~300 tokens)

WHY PARAMETERS MATTER? (Same Prompt – Different Parameters)

Prompt: "Explain AI"			
Temperature	Max Tokens	Result (Example Output)	What you get
0.2	150	"AI is the simulation of human intelligence in machines..."	Short, factual, consistent
0.8	300	"AI is a field that enables machines to learn, reason, and solve problems like humans..."	Balanced, informative and creative
1.2	600	"AI is the art of creating intelligent machines that can imagine, learn from data, make decisions, and even improve themselves..."	Very creative, more diverse and longer

KEY TAKEAWAY

When calling an LLM API, you always send: 1) The **Prompt** + 2) **Parameters** → These parameters control creativity, length, style, and quality of the response.

When we call an API, we don't just send the prompt.

We also send parameters.

Example

```
response = model.generate_content(  
    prompt="Explain AI",  
    temperature=0.8,  
    max_tokens=300  
)
```

These parameters control the response.

Step 5: Temperature Parameter

STEP 5: TEMPERATURE PARAMETER

This is the **MOST important** parameter.



Temperature controls
Randomness
Think of it like creativity.

Low temperature

↓
Less creative

↓
More factual

High temperature

↓
More creative

↓
More diverse answers

TEMPERATURE RANGE

Usually
0.0 to 2.0

Most APIs use
0 to 1



Higher values = more randomness and creativity
Lower values = more accuracy and consistency

TEMPERATURE = 0



0
Very Low

- Model becomes very deterministic.
- Gives the same answer every time.
- Very consistent and factual.

Example Prompt: Explain Python.



Ask 1

Python is a high-level programming language...



Ask again

Python is a high-level programming language...



Again

Python is a high-level programming language...

Output is exactly the same every time.

Use Cases

- Medical
 - Legal
 - Research
 - Banking
 - Coding
- Anywhere accuracy matters.

TEMPERATURE = 0.3



0.3
Low

- Still accurate.
- Slight variation in wording.
- Almost the same meaning.

Example Prompt: Explain Python.



Ask 1

Python is an interpreted programming language...



Next time

Python is one of the most popular programming languages...

Answers are very similar with small changes.

Good for:

- Professional writing
- Reports
- Summaries
- General Q&A

TEMPERATURE = 0.7



0.7
Medium (Default)

- Balanced.
- Good accuracy.
- Good creativity.
- This is the default for many applications.

Example Prompt: Explain Python.



Python is a popular high-level programming language known for its simple syntax and readability. It supports multiple programming paradigms and is widely used in web development, data science, AI, automation, and more...

Good for:

- Chatbots
- Assistants
- Content Generation
- General Applications

TEMPERATURE = 1



1
High

- Creative.
- Responses vary every time.
- More diverse answers.

Example Prompt: Write a slogan for an AI company.



Response 1

Empowering Tomorrow with AI



Response 2

Where Intelligence Meets Innovation



Response 3

Building the Future One Algorithm at a Time

Different every time!

Good for:

- Marketing
- Story Writing
- Brainstorming
- Creative Content

TEMPERATURE = 2



2
Very High

- Very random.
- Sometimes funny.
- Sometimes incorrect.
- Sometimes nonsense.

Example Prompt: Write a slogan for an AI company.



Response 1

AI: Because Humans Need a Brain Upgrade! 🤖



Response 2

Let AI Do the Thinking, You Do the Chill 🧊



Response 3

I Don't Think, Therefore AI Is. 🤔

Answers can be unpredictable or even wrong.

Use with Caution!



High randomness can lead to irrelevant or incorrect answers.

QUICK SUMMARY TABLE

Temperature	Level	Behavior	Best Used For
0	Exact	Deterministic, same answer every time	Medical, Legal, Research, Banking, Coding
0.2	Very Accurate	Very accurate, minimal variation	Reports, Documents, Professional use
0.5	Balanced	Balanced accuracy and creativity	Q&A, Summaries, General content
0.7	Creative + Accurate	Good creativity with good accuracy	Chatbots, Assistants, Default setting
1.0	Creative	Very creative, diverse responses	Marketing, Writing, Slogans, Ideas
2.0	Highly Random	Highly random, may be wrong	Experimental, Fun, Creative exploration

KEY TAKEAWAYS

- ✓ Temperature controls randomness (creativity).
- ✓ Lower values = more factual and consistent.
- ✓ Higher values = more creative and diverse.
- ✓ Most APIs use range 0 to 1.
- ✓ Choose temperature based on your use case.



PRO TIP:

Start with 0.7 (balanced).
Lower it for accuracy.
Increase it for creativity.



This is the MOST important parameter.

Temperature controls

Randomness

Think of it like creativity.

Low temperature

↓

Less creative

↓

More factual

High temperature

↓

More creative

↓

More diverse answers

Temperature Range

Usually

0.0
to
2.0

Most APIs use

0 to 1

Temperature = 0

Model becomes very deterministic.

Ask

Explain Python.

Output

Python is a high-level programming language...

Ask again

Same answer.

Again

Same answer.

Very consistent.

Use Cases

Medical

Legal

Research

Banking

Coding

Anywhere accuracy matters.

Temperature = 0.3

Still accurate

Slight variation

Python is an interpreted programming language...

Next time

Python is one of the most popular programming languages...

Almost same.

Temperature = 0.7

Balanced

Good accuracy

Good creativity

This is the default for many applications.

Temperature = 1

Creative

Responses vary every time.

Example

Prompt

Write a slogan for an AI company.

Response 1

Empowering Tomorrow with AI

Response 2

Where Intelligence Meets Innovation

Response 3

Building the Future One Algorithm at a Time

Different every time.

Temperature = 2

Very random

Sometimes funny

Sometimes incorrect

Sometimes nonsense.

Summary

Temperature	Behavior
0	Exact
0.2	Very Accurate
0.5	Balanced
0.7	Creative + Accurate
1.0	Creative
2.0	Highly Random

Step 6: Maximum Tokens

STEP 6: MAXIMUM TOKENS



This parameter controls **HOW LONG THE RESPONSE CAN BE.**
Not words.
TOKENS.

In simple words:
Maximum Tokens = The limit on the number of pieces of text the model can generate.
When it reaches the limit, it stops.



WHAT IS A TOKEN?

A token is a **small piece of text.**

It can be:

- A word
- Part of a word
- A punctuation
- A number
- A space



Different models and languages may create tokens differently.

EXAMPLE OF TOKENIZATION

Sentence:

Machine Learning is amazing.

May become (tokens):

Machine Learning is amazing .

Around 5 tokens.

Approximately
1 token ≈ 0.75 English word

HOW MAXIMUM TOKENS AFFECT THE RESPONSE

Prompt (same for all):

Explain Deep Learning.

Model (LLM)

The model generates tokens up to the limit you set (Maximum Tokens).



MAXIMUM TOKENS = 20

Very Low Limit

Response:

Deep Learning is a subset of machine learning that uses neural...

(Stops. Incomplete.)

Why?

The model reached the limit (20 tokens) so it had to stop in the middle.

- Short answer
- May not complete the idea

MAXIMUM TOKENS = 100

Medium Limit

Response:

Deep Learning is a subset of machine learning that uses neural networks with multiple hidden layers to learn patterns from large amounts of data. It is inspired by the structure and function of the human brain and is widely used in image recognition, natural language ...

(Better.)

Why?

The model can generate more tokens, so the answer is longer and more complete.

- Covers the main idea
- Still not very detailed

MAXIMUM TOKENS = 500

High Limit

Response:

Deep Learning is a subset of machine learning that uses artificial neural networks with multiple hidden layers to learn complex patterns from large amounts of data. These models automatically learn hierarchical representations of features, making them very powerful for tasks such as image recognition, speech recognition, natural language processing, autonomous driving, medical diagnosis, and many more. Deep learning algorithms include Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Transformers, and more...

(Very detailed explanation.)

Why?

The model has enough token limit to explain in detail.

- Detailed answer
- Covers many points
- May include examples, pros/cons, applications, etc.

SUMMARY

Small Max Tokens

Short answer
Incomplete

Large Max Tokens

Detailed answer
More complete

Choose the right limit based on your need!

IMPORTANT TIPS

- ✓ More tokens = Higher cost (API usage).
- ✓ Too many tokens may include unnecessary information.
- ✓ Too few tokens may cut the answer in the middle.
- ✓ Use a balanced value for good and complete responses.

COMMON RANGE USED

Most APIs allow up to 4,096 tokens, 8,192 tokens, or even 32,768 tokens depending on the model.



This parameter controls

How long the response can be

Not words.

Tokens.

What is a Token?

A token is a small piece of text.

Example

Sentence

Machine Learning is amazing.

May become

Machine

Learning

is

amazing

Around 5 tokens.

Approximately

1 token $\approx$ 0.75 English word

Example

Maximum Tokens = 20

Prompt

Explain Deep Learning.

Response

Deep Learning is a subset of machine learning that uses neural...

Stops.

Incomplete.

Maximum Tokens = 100

Deep Learning is a subset of machine learning that uses neural networks with multiple hidden layers...

Better.

Maximum Tokens = 500

Very detailed explanation.

Summary

Small Max Tokens

↓

Short answer

Large Max Tokens

↓

Detailed answer

Step 7: What Students Must Experiment

They should fix one prompt.

Example

Explain Artificial Intelligence.

Then change only temperature.

Like this

Experiment	Temperature	Max Tokens
1	0	100
2	0.3	100
3	0.7	100
4	1	100

Observe

- Creativity
- Accuracy
- Style

Now keep temperature fixed.

Temperature = 0.7

Change Max Tokens.

Experiment	Temperature	Max Tokens
1	0.7	50
2	0.7	100
3	0.7	300
4	0.7	600

Observe

Response length.

Step 8: What should students analyze?

Response Quality

Is it accurate?

Creativity

Does it generate unique ideas?

Length

Short

Medium

Long

Repetition

Higher temperatures may repeat ideas.

Hallucination

Sometimes LLM invents facts.

High temperature increases this risk.

Example

Prompt

Who invented Artificial Intelligence?

Temperature = 2

Model may invent a person.

Wrong.

Step 9: Sample Experiment

Prompt

Write a birthday wish.

Temperature = 0

Happy Birthday. Wishing you good health and happiness.

Temperature = 0.7

Happy Birthday! May your life be filled with joy, laughter and unforgettable memories.

Temperature = 1

May every sunrise bring new adventures and every sunset leave beautiful memories. Happy Birthday!

Same prompt

Different creativity.

Step 10: Application Flow

Start



User enters prompt



Choose Temperature



Choose Maximum Tokens



Click Generate



Application sends API request



LLM processes prompt



Generated response received



Display output



End

Step 11: Evaluation Table

SUMMARY: IMPACT OF KEY LLM PARAMETERS			
 LLM responses change based on parameter values. Understanding these helps us get the right type and length of answer.			
Parameter	Low Value	High Value	Impact
 Temperature (Controls randomness / creativity)	More factual and consistent  "Paris is the capital of France. It is known for the Eiffel Tower." Output is similar every time.  Consistent & predictable	More creative but less predictable  "Paris, the heart of France, where the Eiffel Tower reaches for the sky and dreams come alive!" Output changes every time.  Varied & creative	 Controls creativity and randomness • Low values = less random, more factual and reliable. • High values = more random, more creative, but less predictable.
 Maximum Tokens (Controls response length)	Short response  "Machine Learning is a field of AI that allows computers to learn from data." Stops early. May be incomplete. 10  50 Fewer tokens used	Long response  "Machine Learning is a field of Artificial Intelligence that allows computers to learn from data, improve from experience, and make predictions or decisions without being explicitly programmed. It uses algorithms, patterns, and statistics to solve complex problems." Gives more details. More complete. 10  50 More tokens used	 Controls response length • Low values = short answers, may miss details. • High values = long answers, more complete and detailed.
 Key Takeaway: • Use Temperature to decide how creative or factual the answer should be. • Use Maximum Tokens to decide how long or detailed the answer should be.		Balance both parameters to get the best quality and length of responses for your application! 	

Step 12: Industry Applications

STEP 12: INDUSTRY APPLICATIONS



LLM parameters like **Temperature** and **Maximum Tokens** are tuned differently depending on the application.

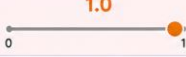
What do these parameters do?



Temperature
Controls randomness
(creativity vs accuracy)



Max Tokens
Controls the maximum
length of the response

Application	Temperature (0 to 1)	Max Tokens	Reason
 Chatbot		300 	 Natural conversation Balanced creativity and coherence for smooth conversations.
 Customer Support		200 	 Accurate and consistent responses Low randomness ensures reliable, fact-based answers.
 Code Generation		500 	 Minimize incorrect code Low randomness + higher token limit helps produce correct and complete code.
 Story Writing		1000 	 Encourage creativity High randomness generates imaginative and diverse stories.
 Email Generation		250 	 Professional yet varied output Moderate creativity for natural and context-appropriate emails.
 Research Summarization		600 	 Detailed and factual summaries Lower randomness for accuracy and higher tokens for comprehensive summaries.
 Key Takeaway: Choose the right combination of Temperature and Max Tokens based on your application goal. There is no one-size-fits-all setting! 			